

Rose Rodrigues

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Education

Bachelor of Science, Computer Science; Mathematics

(December 2026)

College of Science and Engineering, University of Minnesota: Twin Cities - GPA: 3.14

- Minor in Astrophysics
- Coursework: Applied Machine Learning, Modern Cryptography, Analysis of Numerical Algorithms, Algorithms & Data Structures, Maths of Quantum Computing, Formal Languages and Automata Theory

Skills

- **Languages:** Rust, C++, C#, T-SQL, Java, C, Python, JavaScript
- **Services:** Azure DevOps, Git, SSMS, Node.JS, Jenkins, MongoDB, GitHub, AWS
- **Software:** Docker, Unreal Engine, MonoGame, Blender, MATLAB
- **Development Environments:** Windows/Mac/Linux, STM, i.MX 8, Raspberry Pi, Arduino, Cloud
- **Foreign Language:** Portuguese (fluent), Italian (literate & conversational)
- **Embedded Toolchains:** Yocto Project, Torizon OS, STM32CubeIDE, Cross (Rust)

Work Experience

Engineering Co-op

(January 2026 - Present)

AtriCure Inc., Minnetonka, MN

- Develop and debug safety-critical embedded software for medical devices deployed in operating rooms across STM32F7 and i.MX 8M Plus platforms written in Rust, C++, Python and JavaScript.
- Outline and program targeted reliability and verification tests for both software and hardware components across multiple concurrent systems with distinct architectures.
- Built a code generation tool in Rust that automatically syncs scripts across multiple targets based on updates to a centralized CSV error table, replacing a manual 3-file, 2-language, 5-15 minute update process with a 3-second automated step triggered on program compilation.
- Refactored our device's audio system to support a priority queue allowing higher-priority alerts to interrupt active playback while blocking lower-priority ones from doing so; restructured calls to spawn async subprocesses rather than blocking the main service thread, making the priority system possible.

Software Development Intern

(May 2025 - December 2025)

National Checking Company, St. Louis Park, MN

- Engineered novel features using C#, ASP.NET, jQuery, and T-SQL across 16 pull requests, implementing features spanning CRON Jobs, REST APIs, LINQ queries, RDBMS embedded trees, and OIDC auth protocols.
- Contributed to the Agile refinement, development and testing process on the Learning Management System team, validating feature implementations with stakeholder peers.

Tier II Planetarium Guide

(September 2023 - June 2025)

Bell Museum, St. Paul, MN

- Operated specialized real-time dome simulation software to deliver live astrophysical visualizations to audiences of up to 120 people
- Promoted to mentor and onboard new guides, provide structured feedback on current and new presentations, and support event planning, demonstrating initiative and taking on a mentorship role within the team.
- Developed strong technical communication through industry-level science communication to audiences of diverse backgrounds, ages and expertise levels.

Project Experience

Embedded Monophonic Note Recognition System

(December 2025 - Present)

- Designing and implementing a Rust-based note recognition system embedded in a Raspberry Pi using a McLeod Pitch Method for accurate monophonic detection with real-time audio processing.
- Applying explicit requirements tracking and planning a custom PCB design, employing end-to-end embedded systems engineering from specification through hardware deployment.

Pitch-Based Real-Time Controller System

(August 2024)

- Built a dual-program communication system transmitting console inputs as stereo audio frequencies where a C++ program converts button and touchscreen inputs to distinct tones and a Rust decoder and interpreter uses FFT to reconstruct them on its host machine with <50ms latency.

Development of Immersive Metaverses applied to Astrobiology Teaching

(January 2020 - May 2022)

Scientific Pre-Initiation Program, Colegio Dante Alighieri, Sao Paulo, SP, Brazil

- Designed and developed a high-immersion VR educational environment in Unreal Engine, creating procedurally generated terrain via Marching Cubes, Perlin noise and fractional Brownian motion algorithms in C++ as the sole programmer, artist, sound designer and voice actress in the project.
- Authored an original design framework, the Immersive Design Document, adapting Game Design Document methodologies to immersive educational environments through narrative and teaching principles.
- Participated in 11 national and international fairs and won 20+ national and international awards including 1st place at top Brazilian science fairs FEBRACE, MOSTRATEC, and FeNaDANTE; 3rd place in Systems Software at Regeneron ISEF 2022; 1st place at MIICA in Malaysia; and an INCOSE honorable mention; competing against as many as ~1,800 students from 63+ countries.